

SERVICE MANUAL

GE Healthcare Coils USA Instruments

Document 2381683-9

Revision 3

GE Signa[®] Excite[™] 3.0T 8-CHANNEL TORSO ARRAY

GE Catalog Part Number: M1385AY



2381683-9
Rev 3
© 2010, General Electric Company,
All Rights Reserved

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14**-day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

Rev. 6 November 12, 2009

LANGUAGE POLICY FOR SERVICE DOCUMENTATION (DOC0620154 REV 1)

ПРЕДУПРЕЖДЕНИЕ (BG)	Това упътване за работа е налично само на английски език. - Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. - Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. - Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.
警告 (ZH-CN)	本维修手册仅提供英文版本。 - 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 - 未详细阅读和完全理解本维修手册之前，不得进行维修。 - 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。
警告 (ZH-HK)	本服務手冊僅提供英文版本。 - 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。 - 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。 - 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。
警告 (ZH-TW)	本維修手冊僅有英文版。 - 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。 - 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。 - 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。
UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for tekniker, operatøren eller patienten.
WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING (EN)	This service manual is available in English only. • If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services. • Do not attempt to service the equipment unless this service manual has been consulted and is understood. • Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles • Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. • Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. • Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. • Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. • Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. • Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. • Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. • Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. • Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. • Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. • Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. • Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά. • Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης. • Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις. • Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. • Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése. • Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. • Ezen figyelmetztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
ADVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fáanleg á ensku. • Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. • Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilið. • Brot á sinna þessari advörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. • Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. • Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. • Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

警告 (JA)	このサービスマニュアルには英語版しかありません。 • サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 • このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 • この警告に従わない場合、サービスを担当される方、操作員あるいは患者電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. • 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. • 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. • 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. • Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. • Neveiciet aprikojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. • Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.
ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. • Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. • Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. • Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. • Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. • Zanimanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. . • Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. • Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. • Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. • Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

TABLE OF CONTENTS

SECTION 1 – INTRODUCTION	8
1-1 Product Identification and Shipping List	8
1-2 Compatibility	9
1-3 Related Documentation	9
1-4 Environmental Requirements	9
1-5 Theory of Operation	9
section 2 – setup and calibration	11
section 2 – setup and calibration	12
2-1 Coil Installation	12
2-1-1 Special Install Notes	12
2-1-2 Configuration	12
2-2 Installation Functional Checks	12
2-3 Periodic Quality Assurance Check	12
Section 3 – Functional Checks	13
3-1 Scanner Verification	13
3-2 Coil Imaging Performance Verification	13
3-2-1 Tools Required	13
3-2-2 Data Collection	13
3-2-3 Test Explanation and Results Interpretation	25
3-3 External Cable Check	25
3-4 PIN Diodes Check	25
3-5 Mechanical Hardware Check	25
3-6 Troubleshooting Tips	26
Section 4 – Maintenance	28
4-1 Coil Care	28
4-2 Special Care Requirements	28
Section 5 – Replacement	29
SECTION 6 – Renewal Parts	30
6-1 Field Replaceable Units	30
6-2 Other Replaceable Accessories	30
SECTION 7 – Appendix	31
7-1 Schematics	31
7-2 Coil Configuration	32
REVISION HISTORY	33

SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

To identify the 8-Channel Torso Array, refer to the coil labels (see *Figure 1*).



Figure 1: Coil label

Figure 2 shows a picture of the 8-Channel Torso Array.



Figure 2: The 8-Channel Torso Array.

SHIPPING LIST – TABLE 1-1

Box #	Part Name	GE Part #	USAI Part #	Qty
1	Coil	2381683-2	100401	1
1	Operator Manual Packet	2422146-500	2422146-500	1
1	Service Manual	2381683-9	780034	1
2	Patient Liner	2381683-5	502548	1
2	Table Pad	2381683-6	150485	1
2	Spacer Pad	2381683-7	150486	1

1-2 Compatibility

This coil is compatible with the Signa[®] Excite[™] 3.0T MR system.

1-3 Related Documentation

Operator's Guide, GE Part Number 2422146.

Signa[®] Excite[™] 3.0T Service Methods CD, 2389160-1.

1-4 Environmental Requirements

Storage Requirements

The coil should be stored in the scanner room.

Dimensions

43.3" W x 15.0" D x 13.0" H (110.0 cm x 38.1 cm x 101.9 cm)

Weight

14.0 lbs (6.4 kg)

1-5 Theory of Operation

The block diagram of the 8-Channel Torso Array is shown in *Figure 3*. *Figure 4* shows the block diagram of the circuit layout. Also refer to the schematic in the Appendix. The receive-only coil is designed to give optimum signal to noise ratio and uniform coverage of the abdominal and pelvic regions, including the liver, spleen and renal arteries. The coil package consists of anterior and posterior sections, which are positioned above and below the region of interest.

The coil is actively decoupled from the RF transmit coil during transmit by means of multiple RF choking circuits or RF chokes. Each coil element has three active decoupling circuit elements and two passive decoupling circuit elements, which can be switched on actively or passively. The passive RF chokes are switched on using small-signal diodes. The small-signal diodes are turned on by the induced RF voltage in the chokes, coupled from the transmit field (passive coupling). Each small-signal diode is turned on when the induced RF voltage reaches about 0.5 volt. The active RF chokes are switched on using PIN diodes. The PIN diodes are turned on by a forward biasing DC current source supplied by the MRI system (active decoupling). When either the PIN diodes or the small-signal diodes are turned on, the RF choking circuits become very high impedance blocks (typically above 3 k-ohm), compared to the other circuit elements (less than 70 ohm). These high impedance elements segregate the coil circuitry into several isolated electrical segments, preventing any current flow in the coil circuit that might be induced by the body coil transmission. In the event that one of the diodes fails, the signal diode will form a short circuit and permanently de-resonate the coil.

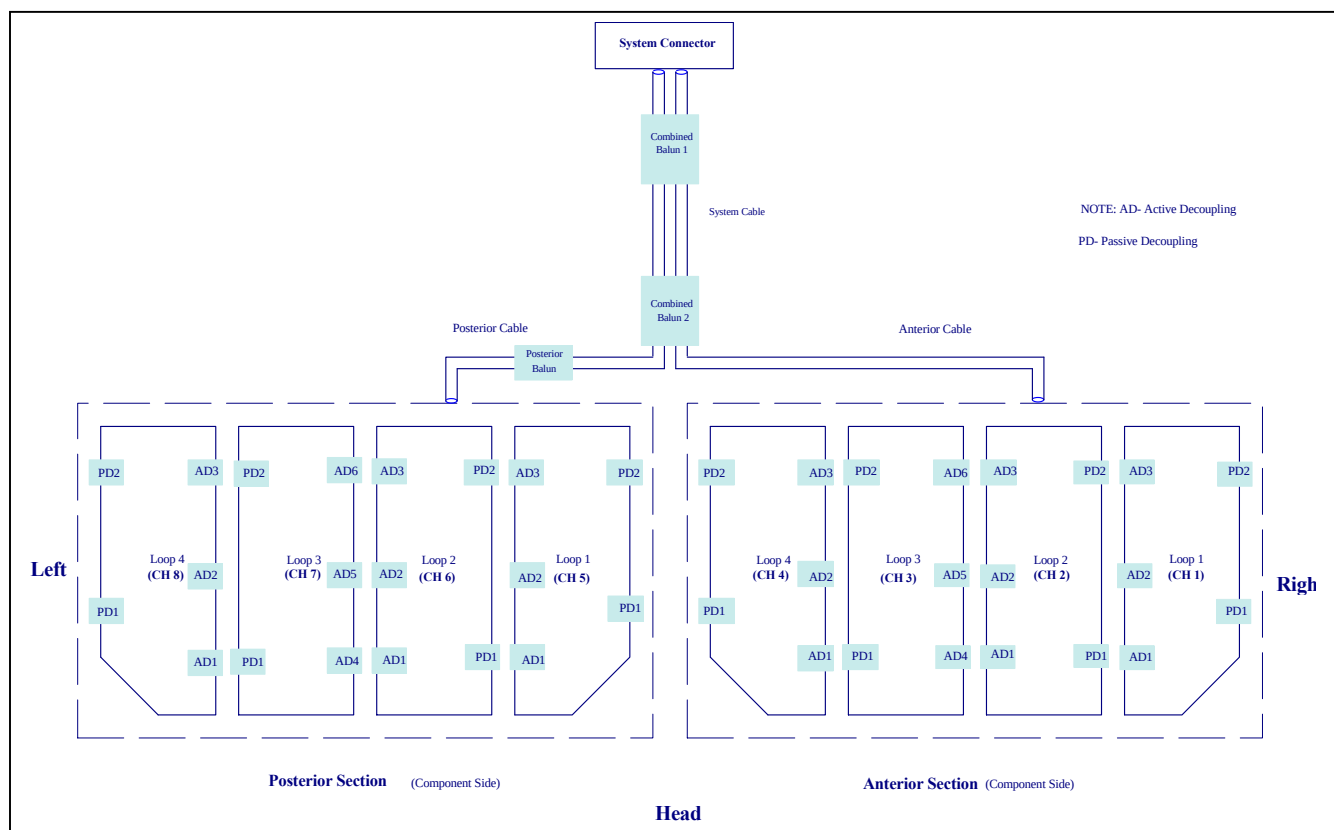


Figure 3: Block diagram of the coil.

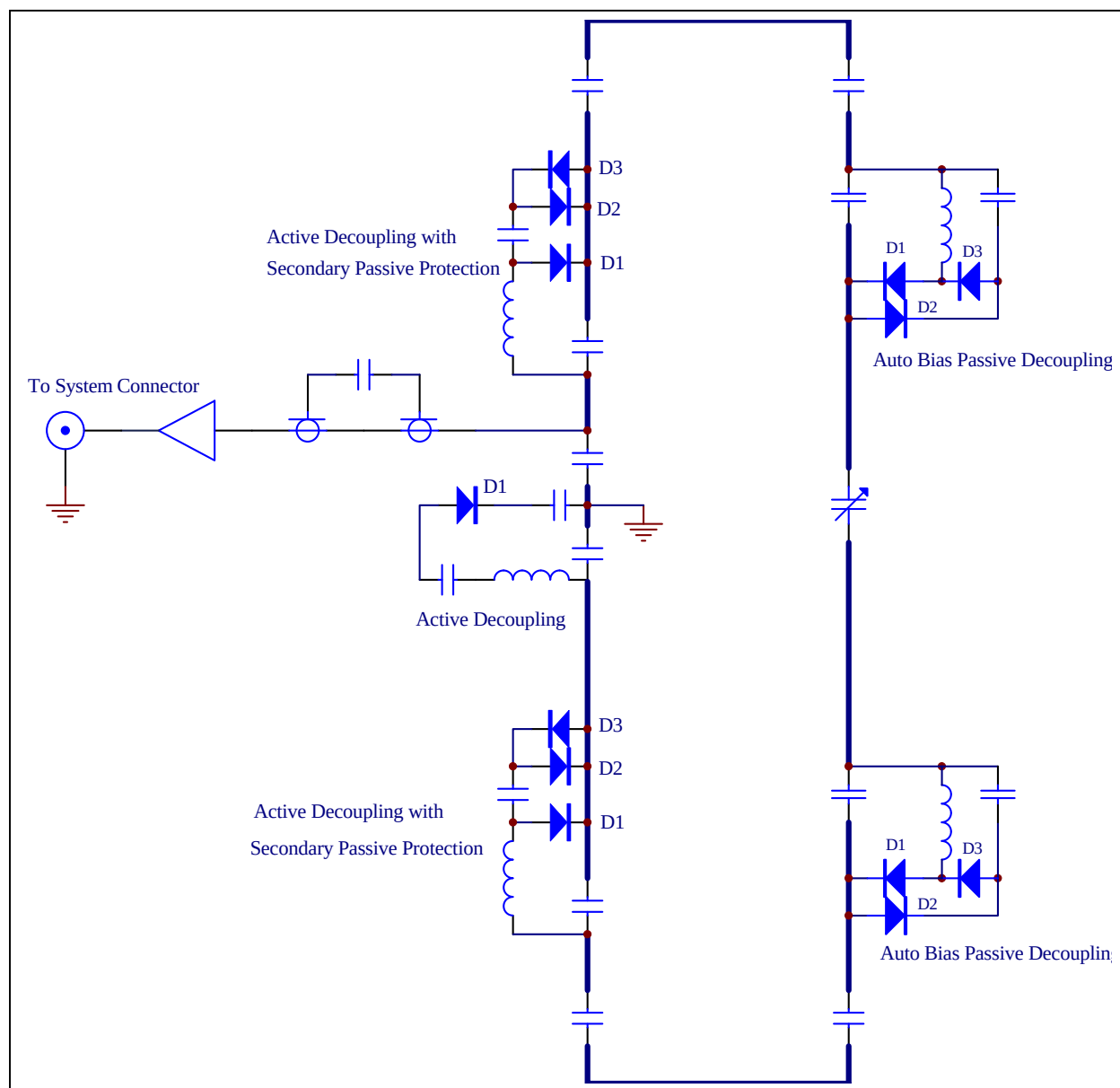


Figure 4: Block diagram of the circuit layout.

SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None.

2-1-2 Configuration

The name for this coil is 8US TORSOPA. Add the coil using the Configuration File Manager. Refer to: Service Methods CD; Software; System Level Software Utilities; Configuration File Manager. For reference, the configuration information is included in the Appendix; however, that configuration is only current as of the printing of this manual.

2-2 Installation Functional Checks

1. Perform system level Signal to Noise Check. Refer to Service Methods CD; Functional Checks; System Level Functional Checks; Signal to Noise Check.
2. Perform Section 3-2 Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2 Coil Imaging Performance Verification.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; Functional Checks; RF Path; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

NOTE: Legacy coils were shipped with a coil specified phantom and positioner. New coils do not ship with a phantom; phantoms come in a unified phantom set with the MR system. New coils do ship with a phantom positioner.

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-1

Description	Qty	Effectivity	GE Part #
Phantom	1	Legacy	2381683-3
Phantom Positioner	1	Legacy	2381683-4
TL Unified Phantom, SiOil	1	Unified	5343347-2

3-2-2 Data Collection

3-2-2-1 Legacy Phantom Procedure

Step 1 – Place the 8-Channel Torso Array on the patient cradle (see *Figure 5*). Note the serial number of the coil.

Step 2 – Position the phantom positioner on the coil as shown in *Figure 6*.



Figure 5 8 Channel Torso Coil Setup



Figure 6 Phantom Positioner Setup

Step 3 – Position the phantom on the phantom positioner as shown in *Figure 7*.

Step 4 – Place the anterior section on the phantom and fasten the Velcro straps on both sides (see *Figure 8*).

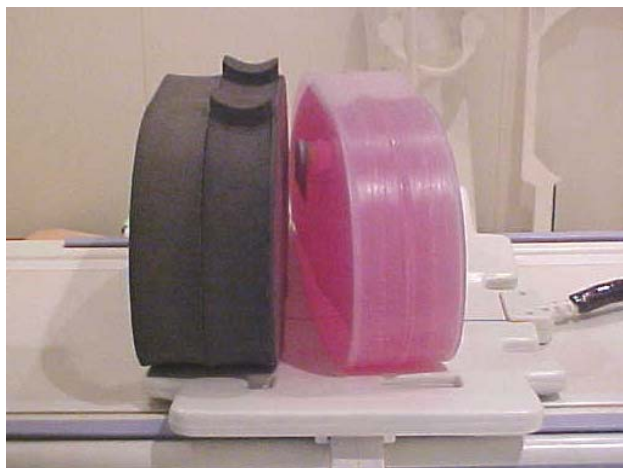


Figure 7 Phantom Setup



Figure Anterior Coil Setup

Step 5 – Establish an axial landmark on the line on top of the coil. Refer to Section 3-2-2-3 to run the MCQA test with the Legacy Phantom.

3-2-2-2 Unified Phantom Procedure



Figure 3-2-2-2-1

Place the posterior section of the coil on the table. (Figure 3-2-2-2-1)

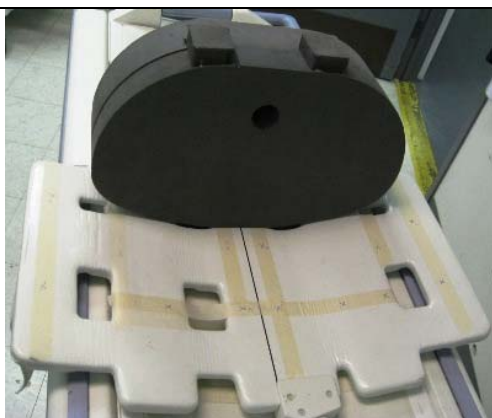


Figure 3-2-2-2-2

Place the positioner of Torso into gaps present on the posterior section of the coil as shown in Figure 3-2-2-2-2



Figure 3-2-2-2-3

Place the TL Unified phantom as shown in figure 3-2-2-2-3

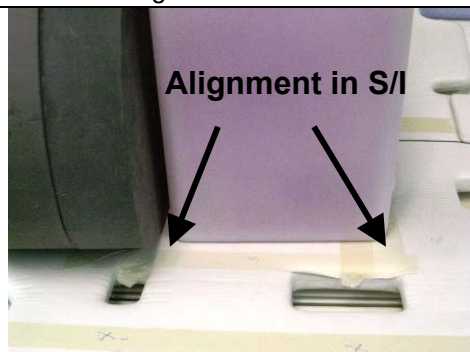


Figure 3-2-2-2-4

Place the TL unified phantom such that the phantom is well centered on both S/I direction as shown in Figure 3-2-2-2-4 and R/L direction as shown in arrows Figure 3-2-2-2-5.

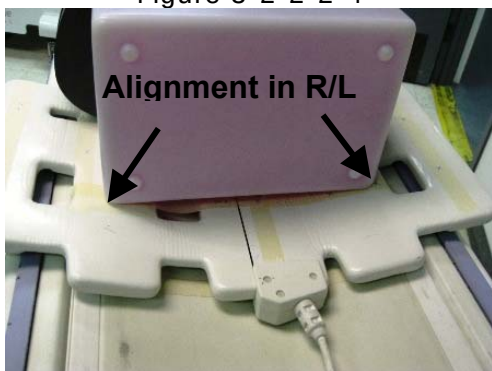


Figure 3-2-2-2-5



Figure 3-2-2-2-6

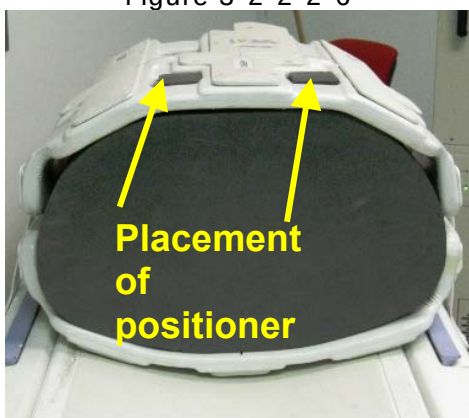


Figure 3-2-2-2-7



Figure 3-2-2-2-8

Place the anterior section (figure 3-2-2-2-6) in such a way that the anterior of the coil is well set in to the positioner as shown in figure 2.3-7 and fasten the Velcro straps so that the coil firmly snugs the phantom (Figure 3-2-2-2-8).



Figure 3-2-2-2-9

Connect the coil to respective port of the system LPCA. Landmark the coil on the cross mark as shown in Figure 3-2-2-2-9 and press advance to scan.

Refer to Section 3-2-2-3 to run the MCQA test with the Unified Phantom Set.

3-2-2-3 Steps to run MCQA

SETUP AND STARTING Multi-Coil Quality Assurance (MCQA) tool USING LEGACY PHANTOM SET

WARNING: All RF coil related tests must be run on a system that is well-calibrated and passing all system tests, especially white pixel, correlated noise, and MCR (Multi-Coil-Receive) tool.

1. If possible, run the Multi-Coil Chain Receive (MCRv) tool to validate receive chain function before running the MCQA tool. If the MCRv tool has missing or failed results, the MCQA tool will detect it and a pop-up will appear, asking if you want to continue.
2. From the Common Service Desktop (CSD) select [Image Quality], *Multi-Coil QA Tool* and Click here to start this tool. The Coil Log Viewer will open as shown in figure 4.1.3-1.

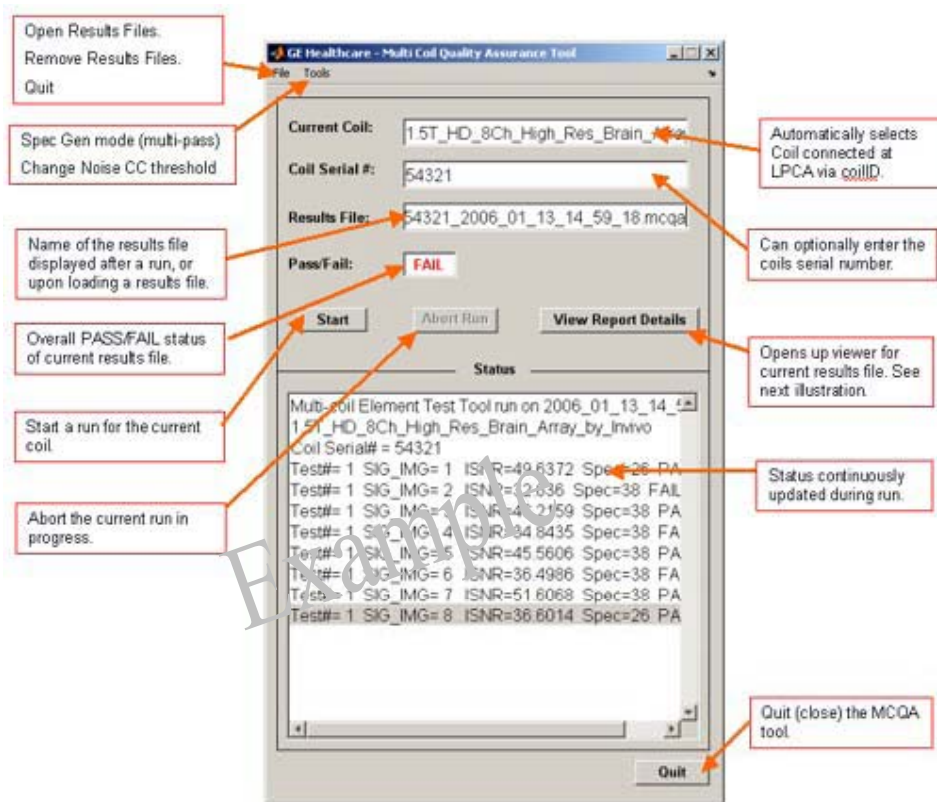


Figure 4.1.3-1

MCQA Starting window

3. The current coil field will be automatically filled in, based on the Coil ID of the coil connected to the LPCA. The names listed will be the physical coil names for each coil.

NOTE: Coil names are specific to the coil connector. Legacy coils should not have reference to the connector, but any new coil with the Bendix connector will have an HDx/HDe identifier in the coil name. During download, the tool should detect if the wrong coil was selected, and give an error. The user will then have the ability to pick again.

4. Enter the *Coil Serial #*: into the provided field.
5. From the MCQA Tool GUI, press the [Start] button to begin the test.
6. A warning dialog (figure 4.1.3-2) will be shown, asking the user to verify the coil/phantom set-up and position. Ensure the phantom and coil setup is proper and select [Yes] to continue.
 - a. The status window of the MCQA Tool GUI will continuously update to give information on what the tool is doing at any point in time.
 - b. A time bar will appear, showing approximate total test time, elapsed time and percent complete, as shown in figure 4.1.3-3.

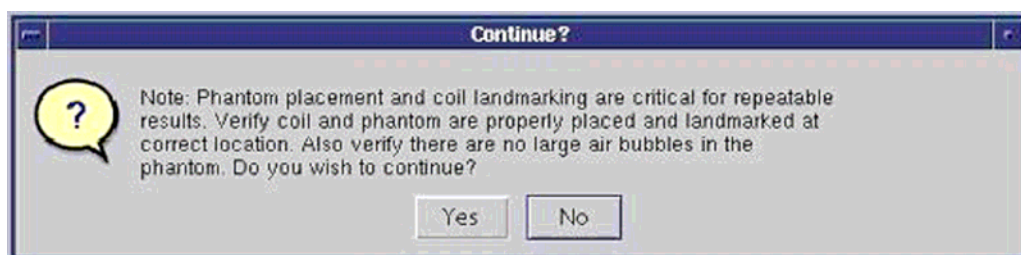


Figure 4.1.3-2

Phantom/Coil, Landmark Warning Popup

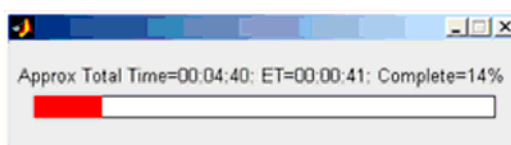


Figure 4.1.3-3

Status and Time Bar

7. Once the test has completed, the name of the Results file will be listed, along with the Pass/Fail status of the test.
8. Record the Measured ISNR and the LSL for each channel displayed in the Status Window on a separate sheet. (See Table 4.1.3-1)

Table 4.1.3. 1: Data Collection Table Example

Si No	Test No.	Element No.	LSL	ISNR

9. Select the [QUIT] button to exit out of the MCQA Tool.
10. Remove the coil and phantom from the system.

Setup and Starting Multi-coil QA (MCQA) Tool USING unified phantom KIT

All RF coil related tests must be run on a system that is well-calibrated and passing all system tests. The system should have passed "Install In Spec" (IIS), especially white pixel, correlated noise, and MCR (Multi-Coil-Receive) tool.

1. From Common Service Desktop (CSD) go to Service Browser and select [Image Quality], Multi-Coil QA Tool and then Click here to start this tool as shown in Figure 4.2.3-1

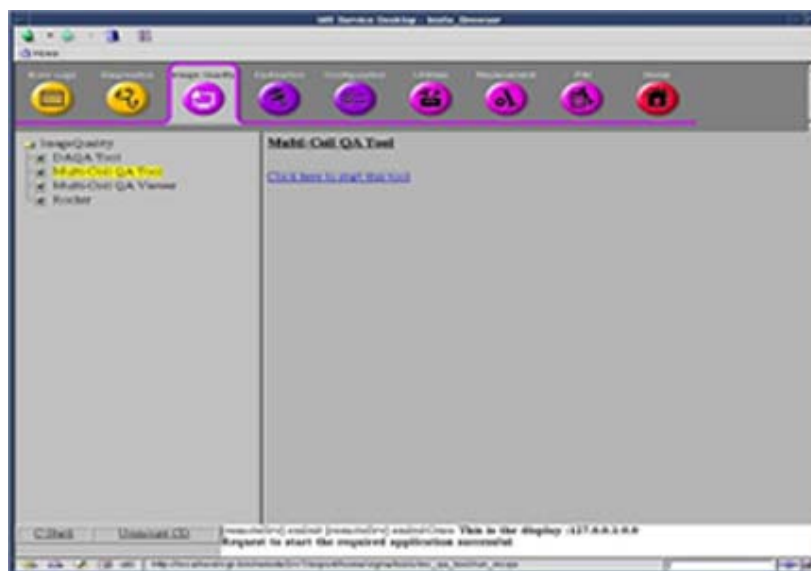


Figure 4.2.3-1

2. Upon start-up, the coil that is connected to the system gets selected in the *Current Coil* field as shown in Figure 4.2.3-2

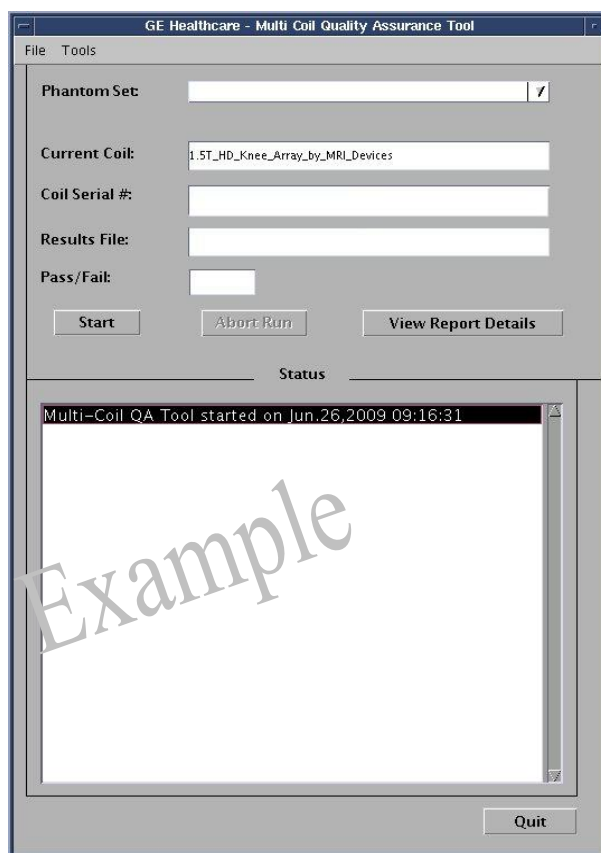


Figure 4.2.3-2

3. A pop-up displays requesting you to, **Please select correct Phantom Set** as shown in Figure 4.2.3-3 below. Click on [OK] to continue.



Figure 4.2.3-3

NOTE: For Optima MR450w, the Unified Phantom Kit is the **ONLY** available phantom set and will be automatically selected in the [Phantom Set] field of the tool.

4. Select the Unified Phantom Set from the drop-down menu as shown in figure 4.2.3-4

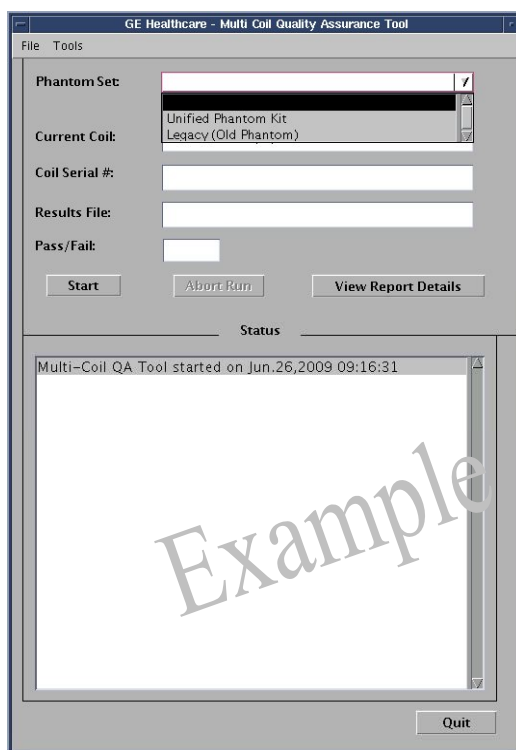


Figure 4.2.3-4

NOTE: For the Legacy Phantom Kit, please select LEGACY (old phantom).

5. The current coil field will be automatically filled in, based on the CoilID of the coil connected to the LPCA. Enter the serial number of the coil being tested in the *Coil Serial Number* field.
6. Click on [Start] to begin the automated test as shown in figure 4.2.3-5. Depending on the number of test locations (complexity of the coil) the test may take from 3 to 20 minutes.

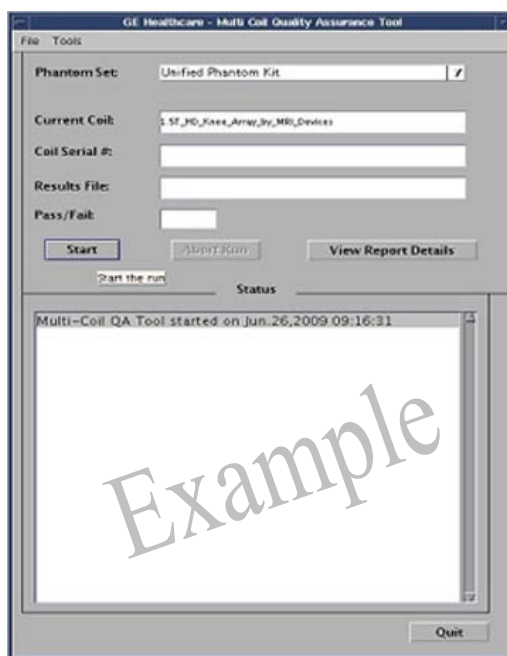


Figure 4.2.3-5

7. A confirmation pop-up displays, asking to continue. Click on [Yes]. See figure 4.2.3-6.



Figure 4.2.3-6

8. Upon start-up, a message stating, **Phantom placement and coil landmarking are critical for repeatable results** will appear. If the landmark has been set correctly and there are no air bubbles in the phantom, click [Yes] to continue. See figure 4.2.3-7.



Figure 4.2.3-7

NOTE: The Status window of the MCQA Tool GUI will continuously update to give information on what the tool is doing at any point in time. A time bar (figure 4.2.3-8) will appear, showing approximate total test time, elapsed time and percent complete. Once the test has completed, the name of the Results file will be listed, along with the Pass/Fail status and the results of the test in the MCQA Tool GUI.

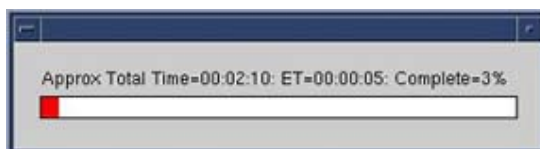


Figure 4.2.3-8

9. When the test is complete, test results display on the screen (figure 4.2.3-9) The PASS/FAIL status shows PASS if all coil elements are functioning properly. If any coil elements display FAIL, call the GE Service Representative for coil repair.

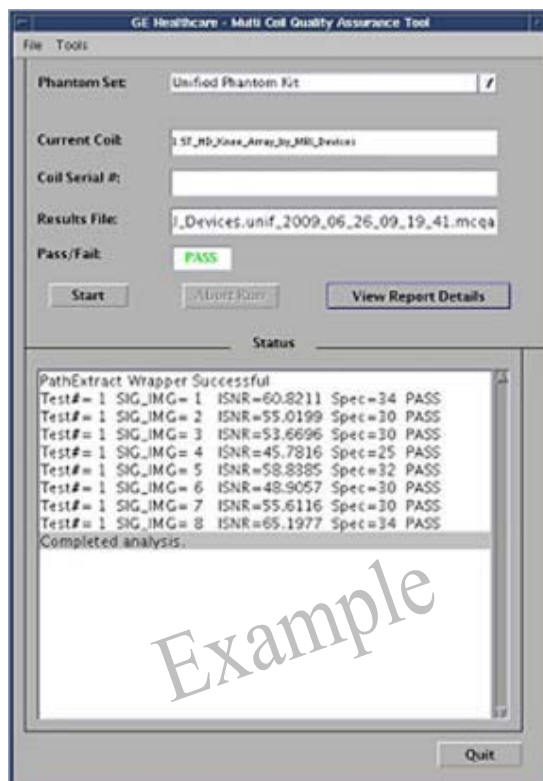


Figure 4.2.3-9

10. The MCQA Tool GUI displays **Fail** for reasons including, but not limited to:
 - a. Bad coil element
 - b. Incorrect phantom used for the test
 - c. Incorrect positioning/placement of the phantom

11. Click on [Quit] button to exit MCQA Tool

Viewing Report Results:

After running the Multi-Coil QA test, select [View Report Details] to review the results. To review past results, select *File, Open Results File* and select the desired coil results file. The Results Viewer will open as shown in *figure 4.3-1*. The results file name and Pass/Fail Results shown on the tool GUI will also be listed across the top of the viewer.

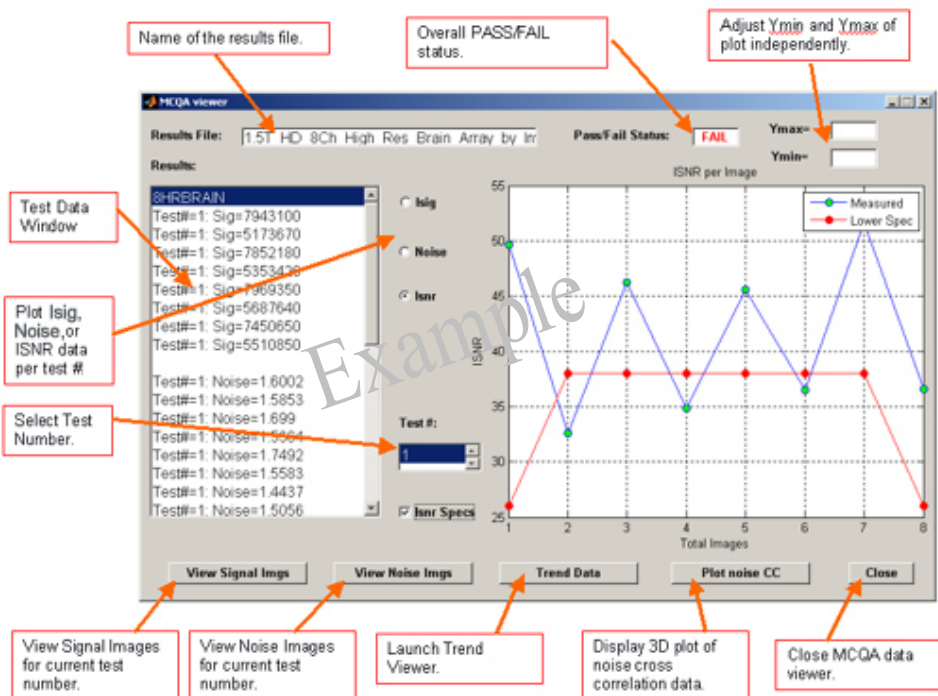


Figure 4.3-1

3-2-3 Test Explanation and Results Interpretation

The 8-Channel Torso Array is a moderately complex phased array. The Quality Assurance tool acquires data looking at the composite image (like the customer sees). Individual element performance is evaluated using the composite image.

Signal images are acquired using a FSE-XL sequence with CV saveinter set to 1 to save intermediate images into the image database. The scan is taken twice for a total of 18 images. Next, a no-RF noise scan is taken, also with saveinter on for a total of 9 noise images. Individual element signal and noise images no longer yield results that are displayed to the user.

The test employs two types of image analyses. Separate signal and noise images are evaluated for individual element signal, noise and SNR. In addition, a filtered version of the NEMA SNR measurement method is also used. The filter takes out adverse effects of subtle ghosting on noise measured in difference.

3-3 External Cable Check

If the signal on any channel is low, refer to *Figure 15* to ensure that none of the B, C or D row pins are bent.

3-4 PIN Diodes Check

No diode checks can be made from the connector. All PIN diode checks must be made individually with the coil open.

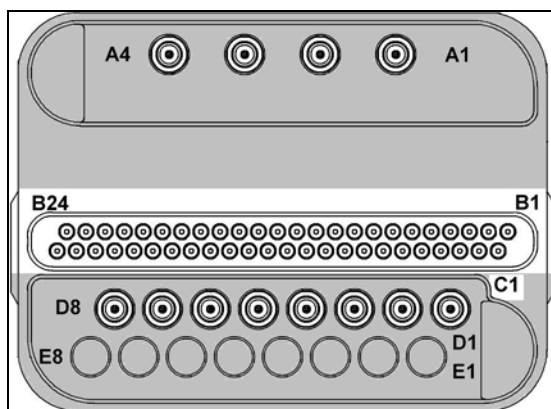


Figure 15: 48-pin Hypertronics connector.

3-5 Mechanical Hardware Check

None.

3-6 Troubleshooting Tips

Symptom #1: The system reports a coil fault during prescan or does not recognize the coil connection to the system when selected in the software.

Probable Cause	Suggested Actions	Resolution
The coil connector has become disconnected from the system interface.	Check to make sure the coil connector is fully engaged.	Engage connector and try the scan again. Make sure the indicator light on the patient table turns green with the proper coil selected on the console.
One or more PIN diodes have failed.	From inside the coil, check the PIN diodes using a DVM. Use diode mode for forward-bias measurements and resistance mode for reverse-bias measurements. Note that there should be no reverse-bias leakage. Bad PIN diodes are usually shorted or exhibit reverse-bias leakage current.	Replace the coil.
There is a DC short or open somewhere within the DC path inside the coil.	Check the coil and associated DC paths with a DVM, in both diode check and resistance modes.	Replace the coil.
There is a problem with the DC bias from the system interface.	Check to ensure the correct DC bias is being provided to the coil during transmit and receive per GEMS System Service Manual. Try a similar 2-channel coil to see if the same problem occurs	Correct MRI system problem.
Coil ID issues.	From CSD, select Diagnostics, Manual Tests, SRI Functional Checks, and run the SRI Coil ID test. Compare the coil ID value found to that listed in Section 7-2 of this manual.	If the value does not match, or no coil ID is found, replace the coil.

Symptom #2: The coil does not pass SNR tests or exhibits poor image quality on patient scans.

Probable Cause	Suggested Actions	Resolution
One or more of the coil channels has a high noise level.	Check QA results from Data Sheet	Replace the coil.
One or more of the coil channels has low signal.	Look at the uncombined images to determine which channel has the problem. Compare signal mean measurements between channels and system performance logs.	Replace the coil.
Excessive ghosting is causing the noise std. dev. measurements to be artificially high.	Window and level the images down to look at the background for signs of ghosting. Try padding the phantom in the holder. Try running the scan without the phantom holder to see if the ghosting diminishes. Try turning off the cold heads to see if the ghosting is vibration-related.	If padding or changing the phantom position can minimize ghosting, then the problem is caused by excessive vibration elsewhere in the system. If the ghosting is not symptomatic of phantom positioning or padding, then replace the coil.
There is a problem with the DC bias from the system interface.	Check to ensure the correct DC bias is being provided to the coils during transmit and receive per GEMS System Service Manual. Check pins for preamp bias.	Correct MRI system DC bias problem – otherwise replace the coil.

SECTION 4 – MAINTENANCE

4-1 Coil Care

WARNING!

Detach the coil connector from the scanner before attempting to clean the coil. Do not touch the connectors with bare fingers. Never press a sharp object against the surface of the connector. Do not reattach the connector after cleaning the coil until the coil has dried completely. Electric shock may result if the coil is attached to the system during cleaning or when it is wet.

CAUTION

Do not spray or pour cleaning solution directly on the coil. Do not submerge the coil in the solution. The coil contains sensitive electronic components that could be damaged by the solution. The coil cannot be sterilized and should be cleaned only according to the procedure outlined in this section.

The following solutions are recommended for the coil and pad surfaces: a ten percent bleach solution (some discoloration may occur) or warm water. Apply cleaning solution to a soft cotton cloth and proceed to clean. To prevent soiling of the coil, the user should place a cotton sheet over the coil before positioning the patient. If the coil is soiled, clean the coil as described above.

4-2 Special Care Requirements

Prior to returning a coil for service, use a ten percent bleach solution (as described above) to eliminate risk of exposure to potentially infectious materials.

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS – TABLE 6-1

Part Name	GE Part #	USAI Part #
Coil	2381683-2	165149
Phantom	2381683-3	155190
Phantom Positioner	2381683-4	155191

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Part Name	GE Part #	USAI Part #
Patient Liner	2381683-5	155192
Table Pad	2381683-6	155193
Spacer Pad	2381683-7	155194
Lower Pad	-	2406572

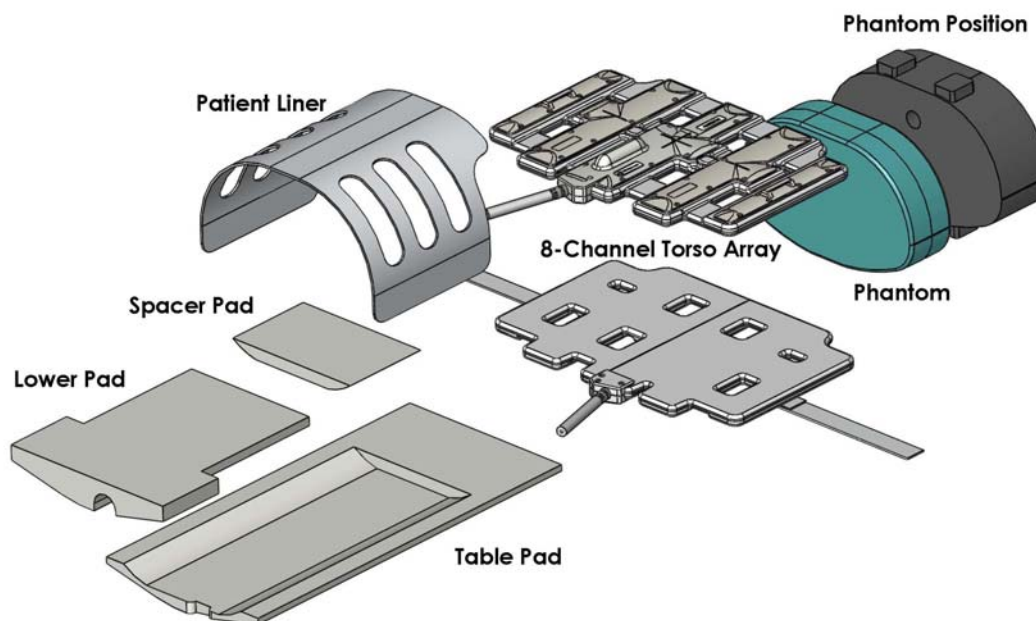
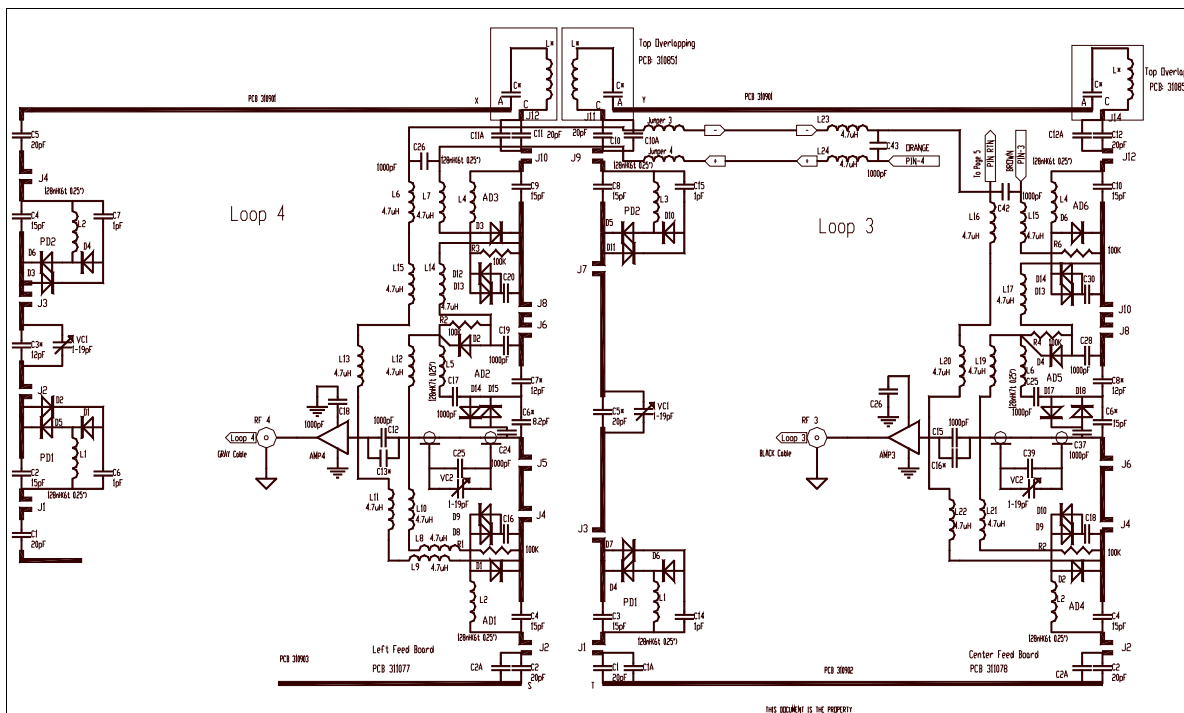


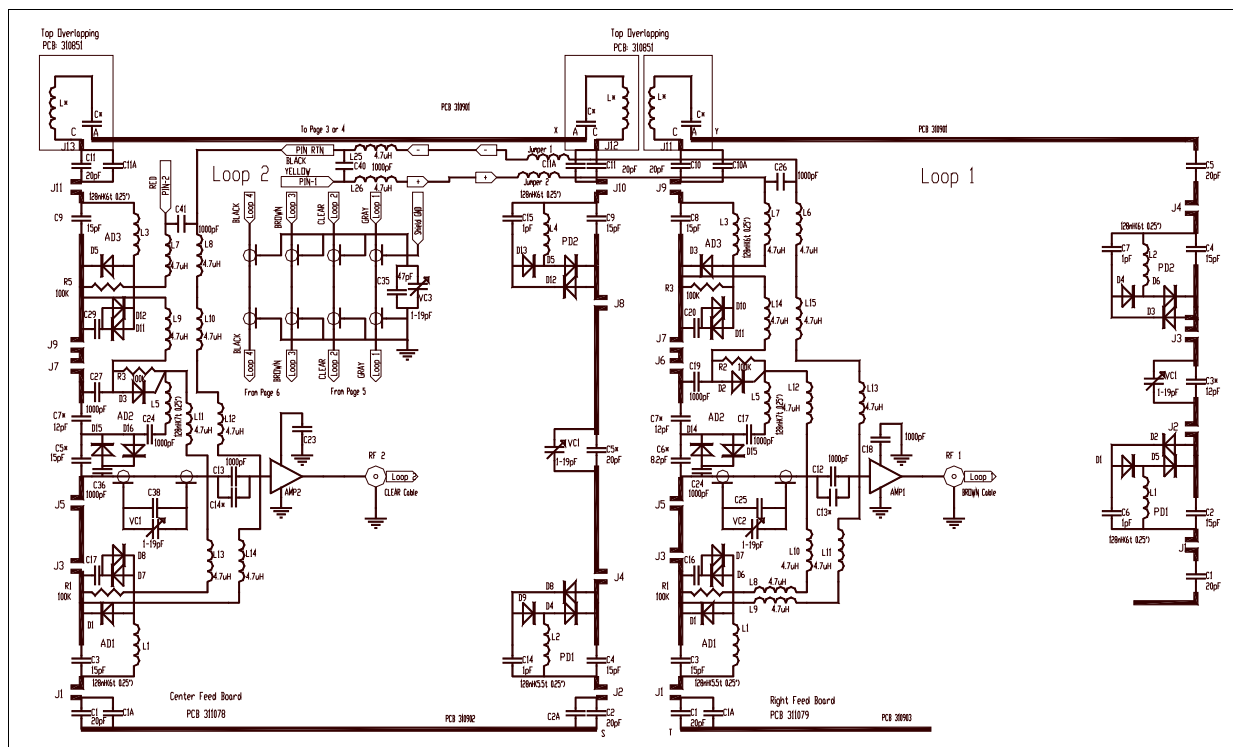
Figure 16: Parts of the coil.

SECTION 7 – APPENDIX

7-1 Schematics



Schematic of Channel 1 and Channel 2



Schematic of Channel 3 and Channel 4

7-2 Coil Configuration

Parameter	Mode
coilName	8US TORSOPA
coilCode	mb156q7appvweb2is03talpsaw63fsn5
physicalCoilDescription	GE by USAI 8 ch torso array
coilType	3
multiCoil	yes
xmitAtten	0
fastTGstartRG	12
fastTGstartTA	90
cableLoss	1.163
coilLoss	0.89
reconScale	0
autoshimRcvr	-1
headDefaultFreqDir	0
mCoilSeperationDir	7
maxFOV	480
assetThresh	10000
nuclide	1H
scic_a	1.5 0.2 0.2 2.0 128 0.4 1.25
scic_s	1.5 0.2 0.2 2.0 128 0.4 1.25
scic_c	1.5 0.2 0.2 2.0 128 0.4 1.25
mcSwitchSel	255
mcBiasEnable	255
mcVoltConfig0	255
mcVoltConfig 1	255
mcOpenCktConfig0	255
mcOpenCktConfig1	0
mcOpenCktConfig2	255
mcOpenCktConfig3	0
mcShortCktConfig0	255
mcShortCktConfig1	0
mcShortCktConfig2	255
mcShortCktConfig3	0
mcChannelSelect	255
startRec	0
endRec	7
numRec	8
userInterfaceGroup	4
prescanWeight0	1.0
prescanWeight1	1.0
prescanWeight2	1.0
prescanWeight3	1.0
prescanWeight4	1.0
prescanWeight5	1.0
prescanWeight6	1.0
prescanWeight7	1.0
aps1_mod	0
aps1_mod_PStloc	0
aps1_mod_PSrloc	0
aps1_mod_fov	200
aps1_plane	1
opthickPS_mod	5
coilPercentExposed	100
optxpos	3
extremity	no
mcReconEnable	0
mcPortEnable	6
mcErrorEnable	6
startFastRec	4
numFastRec	0
endFastRec	4
linearQuad	1

REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
1	10/03	Teresa DeMarco	First Issue
2	07/04	C. Cohara	Sec. 7, Added FRU Lower Pad
3	11Jan10	D. Jackson	Updated damage in transportation and language policy. Updated shipping contents. Updated title page and all headers and footers. Updated operator manual packet number in shipping list and operator manual number in section 1-3. Added new MCQA procedure for Unified phantom set per the Phantom Elimination Program.